

GIRLS WHO CODE

A Beginner's Guide to Computer Science
for Girls Ages 9-14

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WHY CODING MATTERS FOR GIRLS

FEMALE CODER ROLE MODEL

Role Model: Grace Hopper invented the first compiler and coined the term 'debugging.' She proved that women belong at the forefront of computer science. Today only 26% of computing jobs are held by women. You can change that.

THE CONCEPT

Coding is not just about computers. It teaches logical thinking, creative problem-solving, persistence through debugging, and the power to build anything you can imagine. Girls who code become women who change the world.

UNPLUGGED ACTIVITY / CHALLENGE

Think of 3 apps you use daily. Someone CODED those. You could build the next one!

Why Coding Matters for Girls - Practice Zone

PUZZLE/CHALLENGE:

Apply computational thinking to solve this: How would a computer approach why coding matters for girls? Break it into the smallest possible steps.

My solution:

REAL-WORLD CONNECTION:

Where do you see why coding matters for girls in apps/games you use?

Why Coding Matters for Girls - Code It!

WRITE YOUR PSEUDOCODE:

(Pseudocode = instructions in plain English before real code)

Remember: Every expert was once a beginner!

Why Coding Matters for Girls - My Coding Journal

What I learned in this chapter:

Questions I still have:

How I will use this skill:

WHAT PROGRAMMERS ACTUALLY DO

FEMALE CODER ROLE MODEL

Role Model: Susan Wojcicki was the 16th Google employee and later ran YouTube. She saw problems and built solutions that billions of people use every day.

THE CONCEPT

Programmers are problem-solvers who speak the language of computers. They build apps, design games, create websites, analyze data, make robots move, and even help doctors save lives. Programming is in EVERY industry.

UNPLUGGED ACTIVITY / CHALLENGE

List 10 things in your daily life that require code to work. You will be surprised how many there are!

What Programmers Actually Do - Practice Zone

PUZZLE/CHALLENGE:

Apply computational thinking to solve this: How would a computer approach what programmers actually do? Break it into the smallest possible steps.

My solution:

REAL-WORLD CONNECTION:

Where do you see what programmers actually do in apps/games you use?

(Pseudocode = instructions in plain English before real code)

Remember: Every expert was once a beginner!

What Programmers Actually Do - My Coding Journal

What I learned in this chapter:

Questions I still have:

How I will use this skill:

THINKING LIKE A CODER (COMPUTATIONAL THINKING)

FEMALE CODER ROLE MODEL

Role Model: Reshma Saujani founded Girls Who Code to close the gender gap in technology. She teaches millions of girls to think computationally.

THE CONCEPT

Computational thinking has 4 parts: decomposition (breaking big problems into small ones), pattern recognition (finding similarities), abstraction (removing unnecessary details), and algorithm design (creating step-by-step solutions).

UNPLUGGED ACTIVITY / CHALLENGE

Practice decomposition: break 'clean your room' into the smallest possible steps. Write at least 15.

Thinking Like a Coder (Computational Thinking) - Practice Zone

PUZZLE/CHALLENGE:

Apply computational thinking to solve this: How would a computer approach thinking like a coder (computational thinking)? Break it into the smallest possible steps.

My solution:

REAL-WORLD CONNECTION:

Where do you see thinking like a coder (computational thinking) in apps/games you use?

WRITE YOUR PSEUDOCODE:

Remember: Every expert was once a beginner!

Thinking Like a Coder (Computational Thinking) - My Coding Journal

What I learned in this chapter:

Questions I still have:

How I will use this skill:

ALGORITHMS IN EVERYDAY LIFE

FEMALE CODER ROLE MODEL

Role Model: Fei-Fei Li is a leading AI researcher who created ImageNet, teaching computers to recognize images. Her algorithms power visual AI worldwide.

THE CONCEPT

An algorithm is a set of step-by-step instructions to solve a problem. Your morning routine is an algorithm. A recipe is an algorithm. Getting dressed involves an algorithm (you would not put socks over shoes!).

UNPLUGGED ACTIVITY / CHALLENGE

Write an algorithm for making a peanut butter sandwich with EVERY step. Now follow it literally (computers cannot guess!).

Algorithms in Everyday Life - Practice Zone

PUZZLE/CHALLENGE:

Apply computational thinking to solve this: How would a computer approach algorithms in everyday life? Break it into the smallest possible steps.

My solution:

REAL-WORLD CONNECTION:

Where do you see algorithms in everyday life in apps/games you use?

(Pseudocode = instructions in plain English before real code)

Remember: Every expert was once a beginner!

Algorithms in Everyday Life - My Coding Journal

What I learned in this chapter:

Questions I still have:

How I will use this skill:

LOOPS AND PATTERNS

FEMALE CODER ROLE MODEL

Role Model: Megan Smith was the US Chief Technology Officer and VP at Google. She used pattern recognition to solve problems at massive scale.

THE CONCEPT

Loops let computers repeat actions without rewriting code. Instead of writing 'draw star' 100 times, you write 'repeat 100 times: draw star.' Loops are everywhere: playlists on repeat, daily alarms, even seasons cycling.

UNPLUGGED ACTIVITY / CHALLENGE

Find 5 examples of loops in your real life. For each, write how many times it repeats and what the repeated action is.

Loops and Patterns - Practice Zone

PUZZLE/CHALLENGE:

Apply computational thinking to solve this: How would a computer approach loops and patterns? Break it into the smallest possible steps.

My solution:

REAL-WORLD CONNECTION:

Where do you see loops and patterns in apps/games you use?

(Pseudocode = instructions in plain English before real code)

Remember: Every expert was once a beginner!

Loops and Patterns - My Coding Journal

What I learned in this chapter:

Questions I still have:

How I will use this skill:

IF/THEN DECISIONS

FEMALE CODER ROLE MODEL

Role Model: Kimberly Bryant founded Black Girls CODE to increase diversity in tech by teaching coding to African-American girls ages 7-17.

THE CONCEPT

Computers make decisions using IF/THEN logic: IF button pressed THEN play sound. IF password correct THEN open account. IF temperature below 32 THEN display frost warning. Every app is built on thousands of these decisions.

UNPLUGGED ACTIVITY / CHALLENGE

Write 10 IF/THEN rules for your daily life. Example: IF alarm rings THEN get out of bed. IF hungry THEN eat snack.

If/Then Decisions - Practice Zone

PUZZLE/CHALLENGE:

Apply computational thinking to solve this: How would a computer approach if/then decisions? Break it into the smallest possible steps.

My solution:

REAL-WORLD CONNECTION:

Where do you see if/then decisions in apps/games you use?

[illegible]

If/Then Decisions - My Coding Journal

What I learned in this chapter:

Questions I still have:

How I will use this skill:

VARIABLES AND DATA

FEMALE CODER ROLE MODEL

Role Model: Radia Perlman invented the spanning tree protocol that makes the internet work. She is sometimes called the Mother of the Internet.

THE CONCEPT

Variables are containers that hold information. Think of them as labeled boxes: `playerName` holds your name, `score` holds your points, `livesLeft` holds how many tries remain. Programs manipulate variables to create dynamic experiences.

UNPLUGGED ACTIVITY / CHALLENGE

Create variables for a game about yourself: `myName`, `myAge`, `favoriteColor`, `highScore`, `level`. Assign values to each.

Variables and Data - Practice Zone

PUZZLE/CHALLENGE:

Apply computational thinking to solve this: How would a computer approach variables and data? Break it into the smallest possible steps.

My solution:

REAL-WORLD CONNECTION:

Where do you see variables and data in apps/games you use?

(Pseudocode = instructions in plain English before real code)

Remember: Every expert was once a beginner!

Variables and Data - My Coding Journal

What I learned in this chapter:

Questions I still have:

How I will use this skill:

DEBUGGING (FINDING AND FIXING)

FEMALE CODER ROLE MODEL

Role Model: Margaret Hamilton led the software team for NASA's Apollo missions. Her code (and debugging skills) helped land humans on the moon.

THE CONCEPT

Bugs are errors in code. Debugging is detective work: finding where the code does something unexpected and fixing it. Every programmer spends more time debugging than writing new code. It is not failure - it is the job!

UNPLUGGED ACTIVITY / CHALLENGE

Find the bug: in the sequence 'put on socks, put on shoes, put on pants, walk outside' - what is wrong? How would you debug it?

Debugging (Finding and Fixing) - Practice Zone

PUZZLE/CHALLENGE:

Apply computational thinking to solve this: How would a computer approach debugging (finding and fixing)? Break it into the smallest possible steps.

My solution:

REAL-WORLD CONNECTION:

Where do you see debugging (finding and fixing) in apps/games you use?

(Pseudocode = instructions in plain English before real code)

Remember: Every expert was once a beginner!

Debugging (Finding and Fixing) - My Coding Journal

What I learned in this chapter:

Questions I still have:

How I will use this skill:

SIMPLE GAME DESIGN

FEMALE CODER ROLE MODEL

Role Model: Kim Swift designed Portal, one of the most celebrated video games ever. She started as a student and her thesis project became a hit game.

THE CONCEPT

Games are built on: game loop (check input, update state, draw screen, repeat), win/lose conditions, player feedback (sounds, visuals), increasing difficulty, and reward systems. Every game from Tetris to Fortnite uses these elements.

UNPLUGGED ACTIVITY / CHALLENGE

Design a simple game on paper: What are the rules? How do you win? How does difficulty increase? What feedback does the player get?

Simple Game Design - Practice Zone

PUZZLE/CHALLENGE:

Apply computational thinking to solve this: How would a computer approach simple game design? Break it into the smallest possible steps.

My solution:

REAL-WORLD CONNECTION:

Where do you see simple game design in apps/games you use?

(Pseudocode = instructions in plain English before real code)

Remember: Every expert was once a beginner!

Simple Game Design - My Coding Journal

What I learned in this chapter:

Questions I still have:

How I will use this skill:

WEB BASICS (HTML/CSS INTRO)

FEMALE CODER ROLE MODEL

Role Model: Limor Fried founded Adafruit Industries, making open-source hardware and teaching millions to build electronics and code.

THE CONCEPT

Websites are built with HTML (the structure/content) and CSS (the style/appearance). HTML uses tags like `<h1>` for headings and `<p>` for paragraphs. CSS changes colors, fonts, sizes, and layouts. Together they create everything you see online.

UNPLUGGED ACTIVITY / CHALLENGE

Draw a webpage design on paper. Label which parts are HTML structure and which are CSS styling (colors, fonts, positions).

Web Basics (HTML/CSS Intro) - Practice Zone

PUZZLE/CHALLENGE:

Apply computational thinking to solve this: How would a computer approach web basics (html/css intro)? Break it into the smallest possible steps.

My solution:

REAL-WORLD CONNECTION:

Where do you see web basics (html/css intro) in apps/games you use?

Web Basics (HTML/CSS Intro) - Code It!

WRITE YOUR PSEUDOCODE:

(Pseudocode = instructions in plain English before real code)

Remember: Every expert was once a beginner!

Web Basics (HTML/CSS Intro) - My Coding Journal

What I learned in this chapter:

Questions I still have:

How I will use this skill:

APP THINKING

FEMALE CODER ROLE MODEL

Role Model: Whitney Wolfe Herd created Bumble, an app that empowered women to make the first move. She identified a problem and coded a solution.

THE CONCEPT

App development starts with a problem: What frustration do people have? Then: How can technology solve it? Great apps are simple, solve one problem well, and are delightful to use. You can design an app before knowing how to code it!

UNPLUGGED ACTIVITY / CHALLENGE

Identify a problem in your daily life. Design an app to solve it: name it, sketch 5 screens, explain how it works.

App Thinking - Practice Zone

PUZZLE/CHALLENGE:

Apply computational thinking to solve this: How would a computer approach app thinking? Break it into the smallest possible steps.

My solution:

REAL-WORLD CONNECTION:

Where do you see app thinking in apps/games you use?

(Pseudocode = instructions in plain English before real code)

Remember: Every expert was once a beginner!

App Thinking - My Coding Journal

What I learned in this chapter:

Questions I still have:

How I will use this skill:

Chapter 12

AI BASICS

FEMALE CODER ROLE MODEL

Role Model: Timnit Gebru is a leading AI ethics researcher who works to ensure artificial intelligence is fair and does not discriminate against women and minorities.

THE CONCEPT

Artificial Intelligence means teaching computers to learn from data instead of following rigid rules. AI recognizes faces in photos, suggests songs you might like, translates languages, and drives cars. It learns from examples like you learn from experience.

UNPLUGGED ACTIVITY / CHALLENGE

Think about AI you interact with daily (Siri, recommendations, filters). How does it know what you want? What data does it learn from?

AI Basics - Practice Zone

PUZZLE/CHALLENGE:

Apply computational thinking to solve this: How would a computer approach ai basics? Break it into the smallest possible steps.

My solution:

REAL-WORLD CONNECTION:

Where do you see ai basics in apps/games you use?

(Pseudocode = instructions in plain English before real code)

Remember: Every expert was once a beginner!

AI Basics - My Coding Journal

What I learned in this chapter:

Questions I still have:

How I will use this skill:

WOMEN IN TECH (ROLE MODELS)

FEMALE CODER ROLE MODEL

Across tech history, women have been pioneers: Ada Lovelace (first programmer), Grace Hopper (compiler inventor), Hedy Lamarr (WiFi predecessor), Katherine Johnson (NASA calculations), and today's leaders at every major tech company.

THE CONCEPT

Women in tech face unique challenges: being outnumbered, imposter syndrome, bias in hiring. But the women who persist change the world. The tech industry NEEDS your perspective, creativity, and problem-solving approach.

UNPLUGGED ACTIVITY / CHALLENGE

Research one woman in tech who inspires you. What specifically about her career path interests you?

Women in Tech (Role Models) - Practice Zone

PUZZLE/CHALLENGE:

Apply computational thinking to solve this: How would a computer approach women in tech (role models)? Break it into the smallest possible steps.

My solution:

REAL-WORLD CONNECTION:

Where do you see women in tech (role models) in apps/games you use?

(Pseudocode = instructions in plain English before real code)

Remember: Every expert was once a beginner!

Women in Tech (Role Models) - My Coding Journal

What I learned in this chapter:

Questions I still have:

How I will use this skill:

CODING CAREERS

FEMALE CODER ROLE MODEL

The tech industry offers incredible career diversity: game developer, cybersecurity analyst, data scientist, web developer, AI researcher, robotics engineer, UX designer, app developer, tech entrepreneur, and many more.

THE CONCEPT

Coding careers offer: high salaries, flexible schedules, remote work options, creative problem-solving, constant learning, and the ability to build things that millions of people use. Starting to learn NOW gives you years of advantage.

UNPLUGGED ACTIVITY / CHALLENGE

Choose 3 coding careers that interest you. Research: what do they do daily? What do they earn? What skills are needed?

Coding Careers - Practice Zone

PUZZLE/CHALLENGE:

Apply computational thinking to solve this: How would a computer approach coding careers? Break it into the smallest possible steps.

My solution:

REAL-WORLD CONNECTION:

Where do you see coding careers in apps/games you use?

(Pseudocode = instructions in plain English before real code)

Remember: Every expert was once a beginner!

Coding Careers - My Coding Journal

What I learned in this chapter:

Questions I still have:

How I will use this skill:

YOUR CODING JOURNEY STARTS NOW

FEMALE CODER ROLE MODEL

This is not the end - it is the beginning. You now understand how coders think and what programming can do. The next step is to actually start coding! Free resources abound: Scratch, Code.org, Khan Academy, freeCodeCamp, and more.

THE CONCEPT

Your 30-day coding challenge: Week 1 - Complete a Scratch project. Week 2 - Try HTML/CSS on freeCodeCamp. Week 3 - Solve puzzles on Code.org. Week 4 - Start learning Python basics. Track your progress and celebrate milestones!

UNPLUGGED ACTIVITY / CHALLENGE

Set up your coding environment today. Choose one free platform and complete your first lesson. You are officially a coder!

Your Coding Journey Starts Now - Practice Zone

PUZZLE/CHALLENGE:

Apply computational thinking to solve this: How would a computer approach your coding journey starts now? Break it into the smallest possible steps.

My solution:

REAL-WORLD CONNECTION:

Where do you see your coding journey starts now in apps/games you use?

(Pseudocode = instructions in plain English before real code)

Remember: Every expert was once a beginner!

Your Coding Journey Starts Now - My Coding Journal

What I learned in this chapter:

Questions I still have:

How I will use this skill:

YOU ARE A CODER NOW.

Build the future. It needs YOUR code.

PERSONAL GROWTH JOURNAL

What resonated with me most:

Actions I will take:

My commitment:

REFLECTION & ACTION STEPS

Key takeaways from this section:

How this applies to my life:

Questions I still have:

PERSONAL GROWTH JOURNAL

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PERSONAL GROWTH JOURNAL

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My commitment:

REFLECTION & ACTION STEPS

Key takeaways from this section:

How this applies to my life:

Questions I still have:

NOTES & REFLECTIONS

Use this space for additional thoughts and ideas:

PERSONAL GROWTH JOURNAL

What resonated with me most:

Actions I will take:

My commitment:

REFLECTION & ACTION STEPS

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PERSONAL GROWTH JOURNAL

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PERSONAL GROWTH JOURNAL

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My commitment:

REFLECTION & ACTION STEPS

Key takeaways from this section:

How this applies to my life:

Questions I still have:

MY ACTION PLAN

Use this space for your thoughts and reflections:

Use this space for your thoughts and reflections:

ADDITIONAL THOUGHTS

Use this space for your thoughts and reflections:

GROWTH JOURNAL

Use this space for your thoughts and reflections:

MY ACTION PLAN

Use this space for your thoughts and reflections:

GROWTH JOURNAL

Use this space for your thoughts and reflections:

DEEPER EXPLORATION

Use this space for your thoughts and reflections:

MY ACTION PLAN

Use this space for your thoughts and reflections:

Use this space for your thoughts and reflections: